

Q6) the following is a snapshot of algorithm used with pipeline processor (3 marks)

**if (ID/EX.MemRead and**

**((ID/EX.RegRt = IF/ID.RegRs) or**

**(ID/EX.RegRt = IF/ID.RegRt)))**

**then stall the pipeline**

- a) This algorithm is used for \_\_\_\_\_ (1 marks)
- b) Explain this algorithm in details (2 marks)

**Q2)** What is "overflow" in the context of a computer processor, and when does it occur during arithmetic operations, give me four examples? (6 marks)



Fraction of total execution time for each class type for processor 1 and processor 2 (8 marks)

Q1) Consider two different implementations of the same instruction set architecture. There are four classes of instructions, A, B, C, and D. The clock rate and CPI of each implementation are given in the following table.

| Processor   | Clock Rate | CPI Class A | CPI Class B | CPI Class C | CPI Class D |
|-------------|------------|-------------|-------------|-------------|-------------|
| Processor 1 | 5GHZ       | 5           | 3           | 3           | 4           |
| Processor 2 | 3GHZ       | 4           | 5           | 5           | 3           |

Based on this information, calculate (12 marks):

- a) Given a program with  $10^6$  instructions divided into classes as follows: 20% class A, 10% class B, 40% class C, and 30% class D, find the execution time for processor 1 and processor 2? (4marks)

time for processor 1 = .....

time for processor 2 = .....